

Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013

**SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING**
**Product Identifier**

Perihal Produk:

Nickel plating solution, electroless, for copper and copper alloys

Product Description:

Nickel plating solution, electroless, for copper and copper alloys

Cat No. :

44215

**Relevant identified uses of the substance or mixture and uses advised against**

Recommended Use

Laboratory chemicals.

Uses advised against

No Information available

**Company**

 Thermo Fisher Scientific Fisher Scientific (M) Sdn Bhd  
 Hap Seng Business Park, Lot 01-03, 01-04 Aras 1 Unity Square,  
 No 12, Persiaran Perusahaan, Seksyen 23, 40300 Shah Alam,  
 Selangor Darul Ehsan, Malaysia.  
 Main line: +60 3-5525 7888

**Supplier**

E-mail address

Enquiry.my@thermofisher.com

Emergency Telephone Number

 Tel: +03-5525 7888  
 CHEMTREC Malaysia 1-800-815-308 (Malay)  
 CHEMTREC Malaysia (Kuala Lumpur) +(60)-327884561 (Malay)

**SECTION 2: HAZARDS IDENTIFICATION**
**Classification of the substance or mixture**

|                                                      |                     |
|------------------------------------------------------|---------------------|
| Acute oral toxicity                                  | Category 4 (H302)   |
| Acute Inhalation Toxicity - Vapors                   | Category 3 (H331)   |
| Skin Corrosion/Irritation                            | Category 2 (H315)   |
| Respiratory Sensitization                            | Category 1 (H334)   |
| Skin Sensitization                                   | Category 1 (H317)   |
| Germ Cell Mutagenicity                               | Category 2 (H341)   |
| Carcinogenicity                                      | Category 1A (H350i) |
| Reproductive Toxicity                                | Category 1B (H360D) |
| Specific target organ toxicity - (repeated exposure) | Category 1 (H372)   |
| Chronic aquatic toxicity                             | Category 2 (H411)   |

**Label Elements**

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**Signal Word**

**Danger**

## Hazard Statements

H302 - Harmful if swallowed  
H331 - Toxic if inhaled  
H315 - Causes skin irritation  
H317 - May cause an allergic skin reaction  
H334 - May cause allergy or asthma symptoms or breathing difficulties if inhaled  
H341 - Suspected of causing genetic defects  
H350i - May cause cancer by inhalation  
H360D - May damage the unborn child  
H372 - Causes damage to organs through prolonged or repeated exposure  
H411 - Toxic to aquatic life with long lasting effects

## Precautionary Statements

### Prevention

P201 - Obtain special instructions before use  
P202 - Do not handle until all safety precautions have been read and understood  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P270 - Do not eat, drink or smoke when using this product  
P271 - Use only outdoors or in a well-ventilated area  
P272 - Contaminated work clothing should not be allowed out of the workplace  
P280 - Wear protective gloves/protective clothing/eye protection/face protection  
P284 - In case of inadequate ventilation wear respiratory protection

### Response

P302 + P352 - IF ON SKIN: Wash with plenty of soap and water  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P311 - Call a POISON CENTER or doctor  
P330 - Rinse mouth  
P362 + P364 - Take off contaminated clothing and wash it before reuse

### Storage

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed  
P405 - Store locked up

### Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

## Other Hazards

This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

| Component           | CAS No    | Weight % |
|---------------------|-----------|----------|
| Water               | 7732-18-5 | 65       |
| Nickel(II) chloride | 7718-54-9 | 20.00    |
| Sodium chloride     | 7647-14-5 | 10       |

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|                                           |            |   |
|-------------------------------------------|------------|---|
| Butanedioic acid, disodium salt           | 150-90-3   | 3 |
| Phosphinic acid, sodium salt, monohydrate | 10039-56-2 | 2 |

## SECTION 4: FIRST AID MEASURES

### Description of first aid measures

#### **General Advice**

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

#### **Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.

#### **Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.

#### **Ingestion**

Do NOT induce vomiting. Call a physician or poison control center immediately.

#### **Inhalation**

Remove to fresh air. If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required.

#### **Self-Protection of the First Aider**

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

### Most important symptoms and effects, both acute and delayed

May cause allergy or asthma symptoms or breathing difficulties if inhaled. May cause allergic skin reaction. Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing.

### Indication of any immediate medical attention and special treatment needed

#### **Notes to Physician**

Treat symptomatically.

## SECTION 5: FIREFIGHTING MEASURES

### Extinguishing media

#### **Suitable Extinguishing Media**

Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

#### **Extinguishing media which must not be used for safety reasons**

No information available.

### Special hazards arising from the substance or mixture

Thermal decomposition can lead to release of irritating gases and vapors.

#### **Hazardous Combustion Products**

None under normal use conditions.

### Advice for fire-fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

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## SECTION 6: ACCIDENTAL RELEASE MEASURES

### Personal Precautions, Protective Equipment and Emergency Procedures

Use personal protective equipment as required. Ensure adequate ventilation. Keep people away from and upwind of spill/leak. Evacuate personnel to safe areas.

### Environmental precautions

Do not flush into surface water or sanitary sewer system.

### Methods and Material for Containment and Cleaning Up

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

### Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### Precautions for Safe Handling

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance.

### Conditions for Safe Storage, Including any Incompatibilities

Keep container tightly closed in a dry and well-ventilated place.

### Specific End Uses

Use in laboratories.

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

| Component           | Malaysia | ACGIH TLV                  | OSHA PEL                             |
|---------------------|----------|----------------------------|--------------------------------------|
| Nickel(II) chloride |          | TWA: 0.1 mg/m <sup>3</sup> | (Vacated) TWA: 0.1 mg/m <sup>3</sup> |

| Component           | European Union | The United Kingdom                                                            | Germany                                                             |
|---------------------|----------------|-------------------------------------------------------------------------------|---------------------------------------------------------------------|
| Nickel(II) chloride |                | STEL: 0.3 mg/m <sup>3</sup> 15 min<br>TWA: 0.1 mg/m <sup>3</sup> 8 hr<br>Skin | TWA: 0.03 mg/m <sup>3</sup> (8 Stunden).<br>AGW - exposure factor 8 |

### Exposure Controls

#### Engineering Measures

Ensure adequate ventilation, especially in confined areas.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Wear safety glasses with side shields (or goggles)

#### Hand Protection

Protective gloves

#### Skin and body protection

Long sleeved clothing

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Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.  
(Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

## Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators

## Recommended Filter type:

Particulates filter conforming to EN 143

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

When RPE is used a face piece Fit Test should be conducted

## Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice

## Environmental exposure controls

Prevent product from entering drains Do not allow material to contaminate ground water system Local authorities should be advised if significant spillages cannot be contained

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

Appearance

Green

Physical State

Liquid

Odor

Odorless

Odor Threshold

No data available

pH

No information available

Melting Point/Range

No data available

Softening Point

No data available

Boiling Point/Range

No information available

Flash Point

No information available

Method - No information available

Evaporation Rate

No data available

Flammability (solid,gas)

Not applicable

Liquid

Explosion Limits

No data available

Vapor Pressure

No data available

Vapor Density

No data available

(Air = 1.0)

Specific Gravity / Density

No data available

Bulk Density

Not applicable

Liquid

Water Solubility

Miscible

Solubility in other solvents

No information available

Partition Coefficient (n-octanol/water)

Component

log Pow

Butanedioic acid, disodium salt

<-0.59

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|                           |                          |
|---------------------------|--------------------------|
| Autoignition Temperature  | No data available        |
| Decomposition Temperature | No data available        |
| Viscosity                 | No data available        |
| Explosive Properties      | No information available |
| Oxidizing Properties      | No information available |

## SECTION 10: STABILITY AND REACTIVITY

### Reactivity

None known, based on information available.

### Chemical Stability

Stable under normal conditions.

### Possibility of Hazardous Reactions

|                          |                               |
|--------------------------|-------------------------------|
| Hazardous Polymerization | No information available.     |
| Hazardous Reactions      | None under normal processing. |

### Conditions to Avoid

None known.

### Incompatible Materials

None known.

### Hazardous Decomposition Products

None under normal use conditions.

## SECTION 11: TOXICOLOGICAL INFORMATION

### Information on Toxicological Effects

#### Product Information

#### (a) acute toxicity;

|            |                                                                  |
|------------|------------------------------------------------------------------|
| Oral       | Category 4                                                       |
| Dermal     | Based on available data, the classification criteria are not met |
| Inhalation | Category 3                                                       |

### Toxicology data for the components

| Component                                 | LD50 Oral                 | LD50 Dermal                   | LC50 Inhalation            |
|-------------------------------------------|---------------------------|-------------------------------|----------------------------|
| Water                                     | -                         | -                             | -                          |
| Nickel(II) chloride                       | LD50 = 175 mg/kg ( Rat )  | -                             | -                          |
| Sodium chloride                           | LD50 = 3550 mg/kg ( Rat ) | LD50 > 10000 mg/kg ( Rabbit ) | LC50 > 42 mg/L ( Rat ) 1 h |
| Phosphinic acid, sodium salt, monohydrate | LD50 = 7640 mg/kg ( Rat ) | -                             | -                          |

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|  |  |  |  |
|--|--|--|--|
|  |  |  |  |
|--|--|--|--|

(b) skin corrosion/irritation; Category 2

(c) serious eye damage/irritation; No data available

(d) respiratory or skin sensitization;

Respiratory

Category 1

Skin

Category 1

May cause sensitization by skin contact

(e) germ cell mutagenicity; Category 2

(f) carcinogenicity; Category 1A

The table below indicates whether each agency has listed any ingredient as a carcinogen

| Component           | EU           | UK | Germany | IARC    |
|---------------------|--------------|----|---------|---------|
| Nickel(II) chloride | Carc Cat. 1A |    | Cat. 1  | Group 1 |

(g) reproductive toxicity; Category 1B

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; Category 1

Route of exposure

Inhalation

Target Organs

Lungs.

(j) aspiration hazard; No data available

**Symptoms / effects, both acute and delayed** Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing.

**Endocrine Disrupting Properties** Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

### Ecotoxicity effects

The product contains following substances which are hazardous for the environment.  
Contains a substance which is: Very toxic to aquatic organisms.

| Component           | Freshwater Fish                                                                                                                                                                                                       | Water Flea                                                                              | Freshwater Algae                                                                                                                     | Microtox |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|----------|
| Nickel(II) chloride | LC50: = 6.9 mg/L, 96h static (Cyprinus carpio)<br>LC50: = 1.3 mg/L, 96h semi-static (Cyprinus carpio)<br>LC50: > 100 mg/L, 96h static (Brachydanio rerio)<br>LC50: 2.83 - 5.99 mg/L, 96h static (Poecilia reticulata) | EC50: = 0.51 mg/L, 48h Static (Daphnia magna)<br>EC50: = 6.68 mg/L, 48h (Daphnia magna) | EC50: 0.0063 - 0.0125 mg/L, 96h static (Pseudokirchneriella subcapitata)<br>EC50: = 0.66 mg/L, 72h (Pseudokirchneriella subcapitata) |          |

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|                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                     |  |  |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|--|--|
|                 | LC50: 29.76 - 43.57 mg/L, 96h semi-static (Poecilia reticulata)<br>LC50: = 9.65 mg/L, 96h flow-through (Poecilia reticulata)<br>LC50: = 25 mg/L, 96h flow-through (Pimephales promelas)<br>LC50: 2.02 - 6.88 mg/L, 96h static (Pimephales promelas)<br>LC50: 1.9 - 4 mg/L, 96h (Pimephales promelas)<br>LC50: 6.63 - 9.15 mg/L, 96h static (Oncorhynchus mykiss)<br>LC50: 6.7 - 9.7 mg/L, 96h flow-through (Oncorhynchus mykiss)<br>LC50: 2.02 - 6.88 mg/L, 96h static (Lepomis macrochirus)<br>LC50: 18.1 - 25.5 mg/L, 96h flow-through (Lepomis macrochirus) |                     |  |  |
| Sodium chloride | Pimephals prome:<br>LC50: 7650 mg/L/96h                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | EC50: 1000 mg/L/48h |  |  |

## Persistence and degradability

### Persistence

Miscible with water, Persistence is unlikely, based on information available.

### Degradation in sewage treatment plant

Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

## Bioaccumulative potential

Bioaccumulation is unlikely

| Component                       | log Pow | Bioconcentration factor (BCF) |
|---------------------------------|---------|-------------------------------|
| Butanedioic acid, disodium salt | <-0.59  | No data available             |

## Mobility in soil

The product is water soluble, and may spread in water systems. Will likely be mobile in the environment due to its water solubility. Highly mobile in soils.

## Endocrine Disruptor Information

This product does not contain any known or suspected endocrine disruptors

## Other adverse effects

No information available

## SECTION 13: DISPOSAL CONSIDERATIONS

### Waste treatment methods

#### Waste from Residues/Unused Products

Waste is classified as hazardous Dispose of in accordance with the European Directives on waste and hazardous waste Dispose of in accordance with local regulations

#### Contaminated Packaging

Dispose of this container to hazardous or special waste collection point.

#### Other Information

Do not flush to sewer Waste codes should be assigned by the user based on the application for which the product was used Do not empty into drains Do not let this chemical enter the environment

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## SECTION 14: TRANSPORT INFORMATION

### IMDG/IMO

UN-No UN3082  
Hazard Class 9  
Packing Group III  
Proper Shipping Name Environmentally hazardous substances, liquid, n.o.s. (Nickel plating solution, electroless, for copper and copper alloys)

### Road and Rail Transport

UN-No UN3082  
Hazard Class 9  
Packing Group III  
Proper Shipping Name Environmentally hazardous substances, liquid, n.o.s. (Nickel plating solution, electroless, for copper and copper alloys)

### IATA

UN-No UN3082  
Hazard Class 9  
Packing Group III  
Proper Shipping Name Environmentally hazardous substances, liquid, n.o.s. (Nickel plating solution, electroless, for copper and copper alloys)

Special Precautions for User No special precautions required

## SECTION 15: REGULATORY INFORMATION

### Safety, health and environmental regulations/legislation specific for the substance or mixture

International Inventories X = listed

| Component                                 | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | IECSC | AICS | KECL     |
|-------------------------------------------|-----------|------|-----|-------|------|------|-------|------|----------|
| Water                                     | 231-791-2 | X    | X   | X     | X    |      | X     | X    | KE-35400 |
| Nickel(II) chloride                       | 231-743-0 | X    | X   | X     | X    | X    | X     | X    | KE-25837 |
| Sodium chloride                           | 231-598-3 | X    | X   | X     | X    | X    | X     | X    | KE-31387 |
| Butanedioic acid, disodium salt           | 205-778-7 | X    | X   | X     | X    | X    | X     | X    | KE-12377 |
| Phosphinic acid, sodium salt, monohydrate | -         | -    | -   | X     | X    | X    | X     | X    | -        |

### National Regulations

Persistent Organic Pollutant This product does not contain any known or suspected substance  
Ozone Depletion Potential This product does not contain any known or suspected substance

## SECTION 16: OTHER INFORMATION

### Legend

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**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**POW** - Partition coefficient Octanol:Water

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

## Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Prepared By**

**Revision Date**

**Revision Summary**

Health, Safety and Environmental Department

31-Mar-2025

Not applicable.

**In accordance with local and national regulations: Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013**

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**